

MARIA ALM, Austria

WMO: 111370

Lat: 47.406N

Lon: 12.898E

Elev: 794

StdP: 92.14

Time Zone: 1.00 (EUC)

Period: 05-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.5	-12.1	-16.6	1.0	-14.2	-13.9	1.2	-11.9	5.9	2.8	5.1	2.3	1.4	120	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	28.9	19.2	27.0	18.3	25.1	17.6	20.1	26.8	19.2	25.2	18.3	23.7	2.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	14.2	22.6	17.1	13.4	21.5	16.3	12.7	20.6	61.0	26.6	57.9	25.4	55.0	24.0	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.2	4.4	3.6	-16.1	32.4	3.3	1.0	-18.4	33.1	-20.4	33.7	-22.2	34.2	-24.6	34.9		
(5)			-16.6	22.0	3.2	1.0	-18.9	22.7	-20.7	23.3	-22.5	23.9	-24.8	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-3.5	-1.3	3.4	8.5	12.5	16.0	17.9	17.2	12.8	8.2	2.2	-2.3
	DBStd	8.37	4.37	4.25	3.78	3.71	3.70	3.93	3.49	3.30	3.43	3.58	4.11	3.73
	HDD10.0	1743	419	317	204	71	19	4	0	1	13	77	235	382
	HDD18.3	3973	677	551	462	295	182	89	51	61	168	313	483	641
	CDD10.0	896	1	0	1	25	97	184	244	224	96	22	2	0
	CDD18.3	84	0	0	0	0	2	18	37	27	1	0	0	0
	CDH23.3	818	0	0	0	3	49	203	323	225	16	0	0	0
	CDH26.7	195	0	0	0	1	7	52	84	51	1	0	0	0
Wind	WSAvg	1.5	1.3	1.4	1.7	1.8	1.7	1.7	1.6	1.4	1.3	1.4	1.3	1.3
Precipitation	PrecAvg	1421	76	75	99	89	136	174	192	182	132	98	88	81
	PrecMax	1604	186	200	226	152	200	273	293	286	244	173	182	232
	PrecMin	1182	7	18	38	15	34	104	87	116	54	22	4	20
	PrecStd	111	45	46	50	40	44	42	52	43	45	41	40	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	9.5	17.4	23.2	27.3	31.0	31.4	30.3	25.4	19.5	16.3	10.1
		MCWB	7.5	4.7	9.3	12.6	16.6	19.3	20.6	20.1	18.2	14.5	9.8	5.2
	2%	DB	5.1	6.8	14.4	20.5	24.4	28.0	29.0	28.1	22.6	17.2	13.3	5.4
		MCWB	3.9	3.6	7.8	11.8	15.2	18.2	19.2	19.4	16.9	12.9	8.9	4.4
	5%	DB	2.5	5.0	12.0	18.4	22.1	25.7	27.0	25.9	20.5	15.4	9.5	3.2
		MCWB	1.6	2.6	6.6	10.9	14.1	17.3	18.8	18.5	15.6	11.9	7.5	2.4
	10%	DB	1.5	3.6	9.3	16.1	19.9	23.2	24.8	23.6	18.5	13.6	7.4	2.0
		MCWB	0.8	1.8	5.2	9.8	13.0	16.4	17.8	17.5	14.6	11.1	6.3	1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	5.1	9.9	13.4	17.6	20.6	21.3	21.4	18.7	15.1	10.5	6.0
		MCDB	10.2	8.5	16.6	21.7	24.8	28.3	29.3	28.6	23.8	18.3	14.9	8.3
	2%	WB	3.6	3.8	8.2	12.2	16.2	19.1	20.0	20.0	17.4	13.4	8.9	4.2
		MCDB	5.0	6.3	13.7	19.5	22.9	26.2	27.0	26.1	22.0	16.3	12.5	5.4
	5%	WB	1.8	2.9	6.9	11.1	14.9	18.1	19.2	19.0	16.2	12.3	7.6	2.6
		MCDB	2.4	4.6	11.3	17.4	20.9	24.0	25.4	24.5	19.5	14.7	9.6	3.3
	10%	WB	0.9	2.0	5.6	10.1	13.6	17.0	18.3	18.0	15.0	11.3	6.3	1.5
		MCDB	1.4	3.2	9.0	15.4	18.8	21.8	23.6	22.9	17.9	13.3	7.4	1.9
Mean Daily Temperature Range	5% DB	MDBR	5.7	7.3	9.4	11.2	11.3	10.6	10.8	10.7	9.2	8.7	6.0	4.7
		MCDBR	6.0	8.8	14.2	16.2	16.3	16.1	15.4	15.0	12.4	10.9	9.9	5.6
		MCWBR	4.8	5.9	8.0	8.1	7.5	6.8	6.7	7.0	6.9	6.9	6.4	4.4
	5% WB	MCDBR	5.8	7.7	13.6	15.2	14.9	14.0	13.8	13.8	11.8	10.1	9.1	5.3
		MCWBR	4.8	5.3	7.7	7.8	7.1	6.1	6.4	6.9	6.8	6.5	6.0	4.2
Clear-Sky Solar Irradiance	taub	0.277	0.306	0.350	0.391	0.405	0.412	0.411	0.404	0.369	0.339	0.307	0.277	
	taud	2.382	2.298	2.249	2.232	2.252	2.277	2.280	2.320	2.393	2.450	2.470	2.440	
	Ebn at Noon	814	855	864	860	860	853	848	839	838	811	772	772	
	Edh at Noon	69	93	115	129	132	129	127	118	100	81	64	59	
All-Sky Solar Radiation	RadAvg	1.30	2.24	3.22	4.10	4.68	5.18	5.27	4.69	3.43	2.40	1.35	1.06	
	RadStd	0.16	0.28	0.38	0.52	0.56	0.54	0.51	0.56	0.46	0.25	0.22	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (87 neighbors)	+0.41	N/A	N/A	N/A	+0.39	+0.38	N/A	N/A	+62	+28

Nomenclature: See separate page